

Project Charter

Project Name

PHYLO — Physics Learning Openware

Project Overview

PHYLO is an open-source educational platform designed to support practical physics education through affordable, reproducible, and accessible laboratory experiences.

The project combines open hardware, open software, educational resources, and interactive web applications into a unified ecosystem that enables teachers and students to perform real physics experiments using locally available materials.

Although the project originates from secondary-school physics education, its architecture is intentionally designed to support future expansion into additional scientific disciplines, educational levels, and collaborative contributions.

Mission

Develop an open educational platform that enables teachers and students to access, build, use, and improve affordable open-source physics laboratories through freely available hardware designs, software, documentation, and learning resources.

Vision

Enable any educational institution in Latin America to reproduce high-quality experimental laboratories using locally available materials through an open, collaborative, and continuously evolving educational platform.

Target Users

Teachers

Educators looking for affordable, practical laboratory experiences supported by documentation, teaching guides, downloadable resources, and interactive tools.

Students

Learners who perform experiments, collect and analyze data, access learning materials, and better understand physics through hands-on activities.

Developers

Contributors interested in improving the platform, developing new laboratory applications, enhancing documentation, or expanding PHYLO with new features.

Educational Institutions

Schools, universities, research groups, and non-profit organizations interested in adopting, adapting, or contributing to the PHYLO ecosystem.

Core Principles

Open Source

All software, hardware designs, and documentation should be openly accessible whenever possible.

Reproducibility

Laboratories should be designed using affordable, locally available materials that can be reproduced by educational institutions with limited resources.

Hands-on Learning

PHYLO prioritizes real experimentation over purely virtual simulations, encouraging students to interact directly with physical phenomena.

Scalability

The platform should be designed to support continuous growth, allowing new laboratories, applications, educational resources, and contributors to be incorporated without major architectural changes.

Initial Scope

The first public version of PHYLO will focus on providing:

- Laboratory documentation
- Teacher manuals
- Student manuals
- Open hardware files
- Arduino firmware
- Laboratory web applications
- Assembly guides
- Downloadable resources

Future versions may incorporate community contributions, user accounts, collaborative features, simulations, analytics, and additional scientific disciplines.

Success Criteria

PHYLO will be considered successful when educators can independently reproduce laboratory experiences using openly available resources and integrate them into classroom teaching with minimal technical barriers.